

YISHUN JUNIOR COLLEGE
Mathematics Department

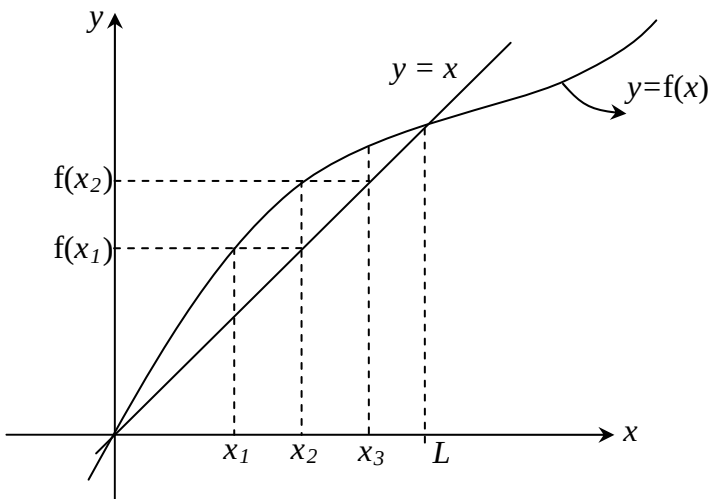
SOLUTION

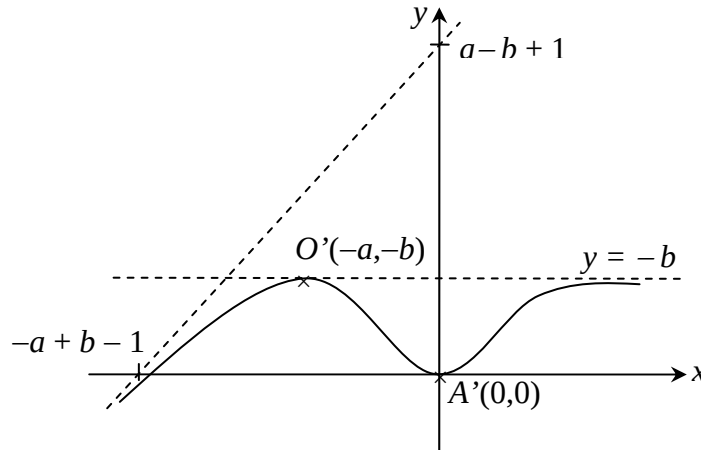
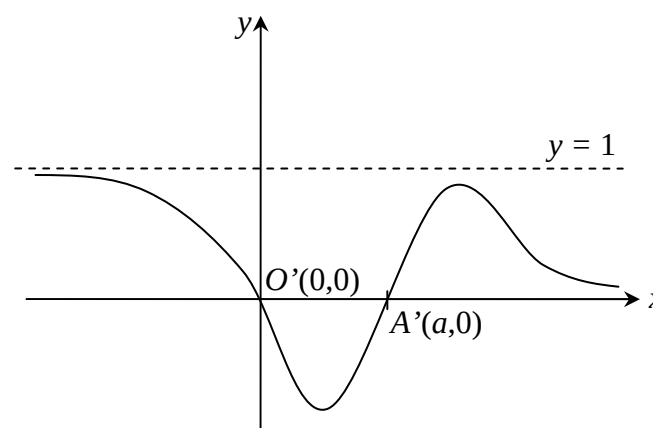
Examination : Preliminary Examination

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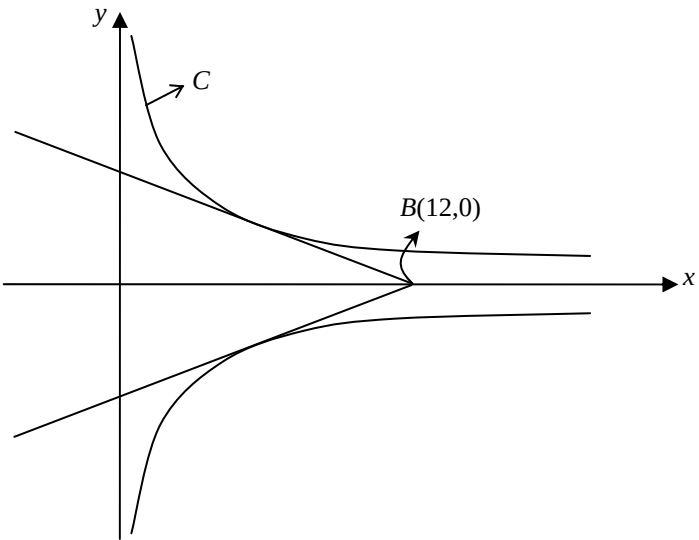
Paper No. : 1

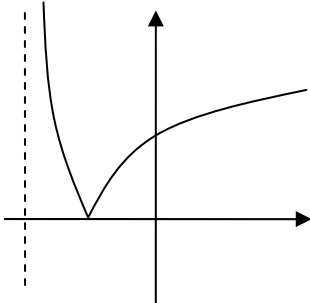
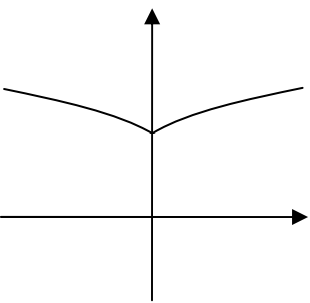
Q ⁿ	Key Steps / Solution
1(a)	Consider $L = \frac{1}{2}L + \frac{L}{e^L}$. Therefore $\frac{1}{2}Le^L - L = 0 \Rightarrow \frac{1}{2}L(e^L - 2) = 0 \Rightarrow L = 0$ or $e^L = 2$ (rejected) $L = \ln 2$ Hence, $L = \ln 2$.
1(b)	
2(i)	$\left(2 + \frac{1}{x}\right)^{-2} = \left(\frac{1}{x}\right)^{-2} (1 + 2x)^{-2}$ $= x^2 (1 - 2(2x) + 3(2x)^2 - 4(2x)^3 + \dots) = x^2 - 4x^3 + 12x^4 - 32x^5 + \dots$ Expansion is valid when $2x < 1$ and $x \neq 0$. That is, $x < \frac{1}{2}$, $x \neq 0$.
2(ii)	Coefficient of $x^r = (-1)^r (r-1)(2)^{r-2}$ or $(-2)^{r-2} (r-1)$
3(i)	$w = \frac{\left \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right ^4}{\left \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right ^5} = \frac{1^4}{1^5} = 1$ $\arg(w) = 4\left(\frac{\pi}{3}\right) + 5\left(\frac{\pi}{6}\right) = \left(\frac{13\pi}{6}\right) \rightarrow \left(\frac{\pi}{6}\right)$ since $-\pi < \arg(w) \leq \pi$ Hence, $\arg(w) = \left(\frac{\pi}{6}\right)$ $w^6 = \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 = \left(\cos \frac{6\pi}{6} + i \sin \frac{6\pi}{6}\right) = -1$

Q ⁿ	Key Steps / Solution
3(ii)	$(\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta)$ $= \cos \theta + \cos^2 \theta - i \cos \theta \sin \theta + i \sin \theta + i \cos \theta \sin \theta + \sin^2 \theta = 1 + \cos \theta + i \sin \theta$ $\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6$ $= \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6$ [Note that $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 = -1$ from part (i)] $= -\left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 = 0$
4(i)	
4(ii)	

Q ⁿ	Key Steps / Solution
5(a)	(i) $\frac{d}{dx} \left(\cos^{-1} \left(\frac{2x}{1+x^2} \right) \right)$ $= \frac{-1}{\sqrt{1 - \left(\frac{2x}{1+x^2} \right)^2}} \left(\frac{(1+x^2)(2) - (2x)(2x)}{(1+x^2)^2} \right)$ $= \frac{-(1+x^2)}{\sqrt{(1+x^2)^2 - 4x^2}} \left(\frac{2+2x^2-4x^2}{(1+x^2)^2} \right)$ $= \frac{-(2-2x^2)}{\sqrt{1-2x^2+x^4}} \left(\frac{1}{1+x^2} \right) = \frac{-2(1-x^2)}{\sqrt{(1-x^2)^2}} \left(\frac{1}{1+x^2} \right)$ Note that $\sqrt{(1-x^2)^2} = \begin{cases} 1-x^2 & \text{if } -1 \leq x \leq 1 \\ -(1-x^2) & \text{if } x < -1 \text{ or } x > 1 \end{cases}$ Since it is given that $-1 \leq x \leq 1$, hence $\frac{d}{dx} \left(\cos^{-1} \left(\frac{2x}{1+x^2} \right) \right) = \frac{-2(1-x^2)}{(1-x^2)} \left(\frac{1}{1+x^2} \right) = \frac{-2}{1+x^2}.$ (ii) From (i), $\int \frac{-2}{1+x^2} dx = \cos^{-1} \left(\frac{2x}{1+x^2} \right) + C''$ But $\int \frac{-2}{1+x^2} dx = -2 \tan^{-1} x + C'$ Hence, $\cos^{-1} \left(\frac{2x}{1+x^2} \right) = -2 \tan^{-1} x + C$ When $x = 0$, $\cos^{-1}(0) = \frac{\pi}{2}$, $-2 \tan^{-1} 0 = 0$ Therefore, $\cos^{-1} \left(\frac{2x}{1+x^2} \right) = -2 \tan^{-1} x + \frac{\pi}{2}$. $\Rightarrow \cos^{-1} \left(\frac{2x}{1+x^2} \right) + 2 \tan^{-1} x = \frac{\pi}{2}$ $\Rightarrow A = 2 \text{ and } B = \frac{\pi}{2}.$ Alternatively : Given $\cos^{-1} \left(\frac{2x}{1+x^2} \right) + A \tan^{-1} x = B \dots (1)$ Therefore $\frac{d}{dx} \left(\cos^{-1} \left(\frac{2x}{1+x^2} \right) + A \tan^{-1} x \right) = 0$ $\Rightarrow \frac{-2}{1+x^2} + \frac{A}{1+x^2} = 0 \Rightarrow \frac{A-2}{1+x^2} = 0 \Rightarrow A-2=0 \Rightarrow A=2$ Next substitute $A = 2$ and $x = 0$ into (1) : $\cos^{-1}(0) + 2 \tan^{-1} 0 = B \Rightarrow B = \frac{\pi}{2}.$

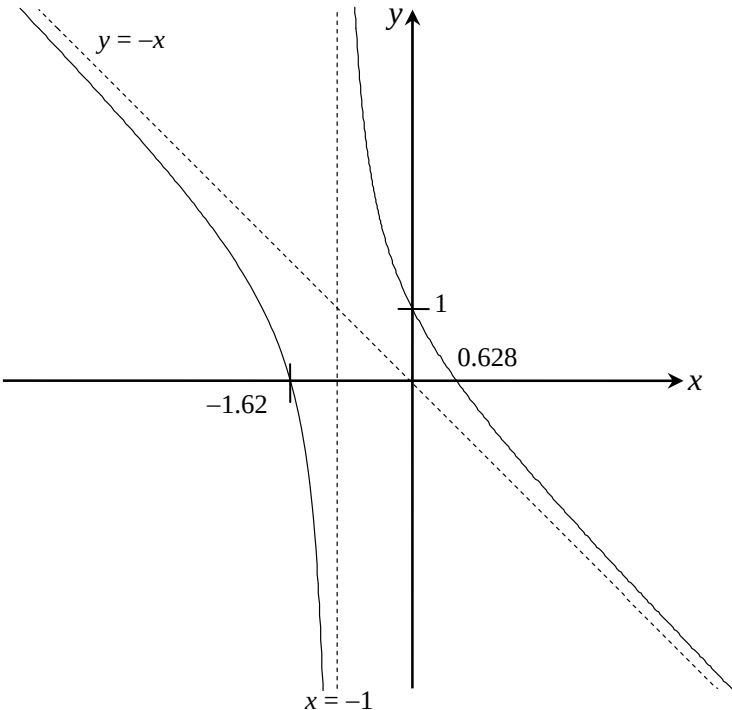
Q ⁿ	Key Steps / Solution
5(b)	Let the volume of the cone by V, and area of the circular base be M. $V = \frac{1}{3}(M)(10) = \frac{10}{3}M \Rightarrow \frac{dV}{dt} = \frac{10}{3} \frac{dM}{dt}$ $\Rightarrow \frac{dV}{dt} = \frac{10}{3}(2) = \frac{20}{3} \text{ cm}^3\text{s}^{-1}.$ Let the radius of the circular base be r. $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$ $V = \frac{1}{3}\pi r^2(10) = \frac{10}{3}\pi r^2 \Rightarrow \frac{dV}{dr} = \frac{20}{3}\pi r$ $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt} = \frac{dV}{dr} \times \frac{dr}{dA} \times \frac{dA}{dt}$ $= \left(\frac{20}{3}\pi r\right) \left(\frac{1}{2\pi r}\right) (2) = \frac{20}{3} \text{ cm}^3\text{s}^{-1}.$ A diagram of a cone. A vertical dashed line from the apex to the center of the base is labeled '10'. A horizontal dashed line from the center of the base to the edge is labeled 'r'.
6(i)	$\frac{dy}{dx} = (2-x)e^{-x} \Rightarrow y = \int (2-x)e^{-x} dx$ $\Rightarrow y = -(2-x)e^{-x} - \int e^{-x} dx + C,$ where C is an arbitrary constant. $\Rightarrow y = (x-2)e^{-x} + e^{-x} + C,$ $\Rightarrow y = xe^{-x} - e^{-x} + C$ When $x = 0, y = 0$. Therefore, $C = 1$. Therefore, $y = xe^{-x} - e^{-x} + 1$. 6(ii) A sketch of $y = xe^{-x} - e^{-x} + 1$ The figure shows a graph of the function y = xe^{-x} - e^{-x} + 1 on a Cartesian coordinate system. The curve starts at the origin (0,0), rises to a peak, and then gradually approaches a horizontal dashed line at y=1 as x increases. A vertical dashed line is drawn from the peak of the curve down to the x-axis at x=2. Max y occurs when $x = 2$. $\frac{dx}{dt} = e^x \Rightarrow \int e^{-x} dx = \int 1 dt \Rightarrow e^{-x} = -t + C.$ $\Rightarrow x = -\ln(-t + C)$ When $t = 0, x = 0$. Therefore $C = 1$. $\therefore x = -\ln(1-t)$ When $x = 2, 2 = -\ln(1-k)$ $\Rightarrow k = 1 - e^{-2}$

Q ⁿ	Key Steps / Solution
7(i)	$\frac{dy}{dt} = \frac{-1}{t^2} \quad \text{and} \quad \frac{dx}{dt} = 2t.$ $\frac{dy}{dx} = -\frac{1}{2t^3}.$ Equation of tangent : $y - \frac{1}{t} = -\frac{1}{2t^3}(x - t^2)$ $\Rightarrow 2t^3y - 3t^2 + x = 0.$
7(ii)	
7(iii)	Substitute $B(12,0)$ into $2t^3y - 3t^2 + x = 0$. $\Rightarrow t^2 = 4 \Rightarrow t = \pm 2.$ When $t = 2$, the point on the curve is $(4, \frac{1}{2})$. When $t = -2$, the point on the curve is $(4, -\frac{1}{2})$. Area of region $R = 2 \times \int_4^a y \, dx - \frac{1}{2} \times (1) \times (12 - 4)$ $= 2 \times \int_2^{\sqrt{a}} \left(\frac{1}{t}\right)(2t) \, dt - 4$ $= 4 \int_2^{\sqrt{a}} 1 \, dt - 4$ $= 4(\sqrt{a} - 2) - 4 = 4\sqrt{a} - 12 \text{ unit}^2.$

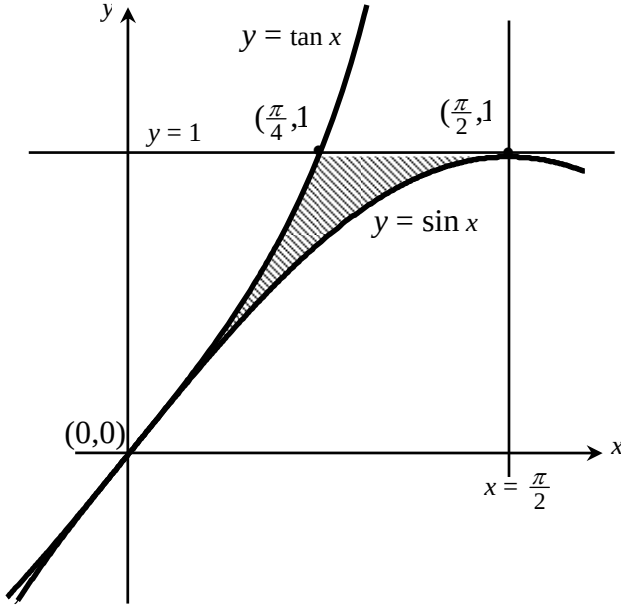
Q ⁿ	Key Steps / Solution
8(a)	$fg(x) = gf(x) \Rightarrow f(\ln(x + 2)) = g(\ln(x + 2))$ $\Rightarrow \ln[\ln(x + 2) + 2] = \ln[\ln(x + 2) + 2]$ $\Rightarrow \ln(x + 2) = \ln(x + 2)$  $y = \ln(x + 2)$  $y = \ln(x + 2)$ Consider $\ln(-x + 2) = -\ln(x + 2)$ $\Rightarrow (-x + 2) = \frac{1}{x + 2}$ $\Rightarrow (2 + x)(2 - x) = 1$ $\Rightarrow 4 - x^2 = 1$ $\Rightarrow x^2 = 3$ $\Rightarrow x = -\sqrt{3}$ (reject $x = \sqrt{3}$) Hence, for $fg(x) = gf(x)$, $x \geq 0$ or $x = -\sqrt{3}$
(b)(i)	$g(x) = \ln(x + 2)$. Note that both $x = 1$ and $x = -1$ are in the domain of g. Clearly $-1 \neq 1$, But $g(-1) = g(1) = \ln 3$. Therefore, function g is not 1-1 and inverse of function of g cannot be formed.
(ii)	largest $a = 0$.
(iii)	Let $y = g(x) = \ln(x + 2)$. Since $x \leq 0$, $y = \ln(-x + 2)$ $\Rightarrow -x + 2 = e^y$ $\Rightarrow x = 2 - e^y$ $\Rightarrow g^{-1}(y) = 2 - e^y$ Hence $g^{-1}: x \rightarrow 2 - e^x$, $x \in [\ln 2, \infty)$

Q ⁿ	Key Steps / Solution
9(a)	l_1 passes through $A(0, 3, 1)$ and is parallel to $m_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$. l_2 passes through $B(1, 0, 2)$ and is parallel to $m_2 = \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$. Since $m_2 = -2m_1$, l_1 and l_2 are parallel or coincident. Since $\vec{AB} = \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \neq km_1, k \in \mathbb{R}$, the line through A and B is not parallel to l_1 and l_2. l_1 and l_2 are distinct parallel lines.
(b)(i)	Since P lies on l_1, ie. $\vec{OP} = \begin{pmatrix} \lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix}$, for some $\lambda \in \mathbb{R}$. $\vec{BP} = \begin{pmatrix} \lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} \lambda-1 \\ 3-2\lambda \\ \lambda-1 \end{pmatrix}$. If APB is a right angle, $\vec{BP} \bullet \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0$. $\begin{pmatrix} \lambda-1 \\ 3-2\lambda \\ \lambda-1 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0$ $(\lambda-1) - 2(3-2\lambda) + (\lambda-1) = 0 \Rightarrow \lambda = \frac{4}{3}$ $\vec{OP} = \begin{pmatrix} \frac{4}{3} \\ 3-2\left(\frac{4}{3}\right) \\ 1+\frac{4}{3} \end{pmatrix} = \begin{pmatrix} \frac{4}{3} \\ \frac{1}{3} \\ \frac{7}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$. Hence the coordinates of P is $\left(\frac{4}{3}, \frac{1}{3}, \frac{7}{3}\right)$
(b)(ii)	Shortest distance between l_1 and l_2 $= \left \vec{BP} \right = \left \begin{pmatrix} \frac{4}{3}-1 \\ 3-2\left(\frac{4}{3}\right) \\ \frac{4}{3}-1 \end{pmatrix} \right = \left \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} \right = \sqrt{\frac{1}{3}}.$

Q ⁿ	Key Steps / Solution
(c)	Normal vector of plane: $\mathbf{n} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \vec{AB} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ Equation of plane : $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = -1$
10(a)	$ \begin{array}{r} \overline{ax + (-1-ac)} \\ x+c \overline{) ax^2 - x + b} \\ \underline{ax^2 + acx} \\ (-1-ac)x + b \\ \underline{(-1-ac)x + (-1-ac)c} \\ b + (1+ac)c \end{array} $ Hence the asymptotes are $x = -c$ and $y = ax + (-1 - ac)$
(b)(i)	When $a = -1$ and $c = 1$, $y = -x + \frac{b}{x+1} \Rightarrow \frac{dy}{dx} = -1 - \frac{b}{(x+1)^2}$ OR $y = \frac{-x^2 - x + b}{x+1} \Rightarrow \frac{dy}{dx} = \frac{(x+1)(-2x-1) - (-x^2 - x + b)(1)}{(x+1)^2}$ $\Rightarrow \frac{dy}{dx} = \frac{-x^2 - 2x - 1 - b}{(x+1)^2}$ Let $\frac{dy}{dx} < 0$. Hence $-x^2 - 2x - 1 - b < 0$ $\Rightarrow$ Discriminant < 0 $\Rightarrow 4 - 4(-1)(-1-b) < 0$ $\Rightarrow 4 - 4(b+1) < 0$ $\Rightarrow b+1 > 1$ $\Rightarrow b > 0$ Alternatively, Let $\frac{dy}{dx} < 0$. Hence, $-1 - \frac{b}{(x+1)^2} < 0$ $\Rightarrow \frac{b}{(x+1)^2} > -1 \Rightarrow b > -(x+1)^2$ Since $(x+1)^2 \geq 0 \quad \forall x \in \mathbb{R}$, therefore $b > 0$. Note that when $b = 0$, $y = \frac{-x^2 - x}{x+1} = \frac{-x(x+1)}{x+1} = -x \text{ also has a negative gradient.}$ Hence, the required range of b is $b \geq 0$.

Q ⁿ	Key Steps / Solution
(b)(ii)	When $a = -1$, $b = 1$ and $c = 1$, $y = \frac{-x^2 - x + 1}{x + 1}$. 
11 (a)(i)	Note that the progression of A is 1, 3, 5, 7, 9, 11, ... (consists of all positive odd numbers) nth term of G, $T_n = \frac{a}{2}(3^n - 1) - \frac{a}{2}(3^{n-1} - 1)$ $= \frac{a}{2}(3^n - 3^{n-1}) = \frac{a}{2}(3^{n-1})(3 - 1) = a(3^{n-1})$ 3^{n-1} is odd for all $n \geq 1$. Hence, for $a(3^{n-1})$ to be a term in A, then $a = 2m - 1$, $m \in \mathbb{Z}^+$. That is, a may be any positive odd integer. (a)(ii) Let the new progression be H : $2^{t_1}, 2^{t_2}, 2^{t_3}, 2^{t_4}, \dots$ Let d be the common difference of A. Consider, $\frac{(n+1)\text{th term of H}}{(n)\text{th term of H}}$ $= \frac{H_{n+1}}{H_n} = \frac{2^{t_{n+1}}}{2^{t_n}} = 2^{t_{n+1} - t_n} = 2^d = 2^2 = 4, \text{ which is a constant.}$ Hence H is a geometric progression. (b)(i) Value of the machine after 5 years $= 70000(0.9)^5$ $= \$41334.30.$ (b)(ii) Maximum revenue that could possibly be generated by the machine $= \frac{14000}{1 - (0.93)} = \$200000.$

Q ⁿ	Key Steps / Solution										
(b)(iii)	Consider $\frac{14000(1 - (0.93)^n)}{1 - (0.93)} + 70000(0.9)^n - 70000 \geq 40000$ $20(1 - (0.93)^n) + 7(0.9)^n - 7 \geq 4$ $20(0.93)^n - 7(0.9)^n \leq 9$ From G.C., <table border="1"> <thead> <tr> <th>n</th><th>a_n</th></tr> </thead> <tbody> <tr> <td>...</td><td>...</td></tr> <tr> <td>5</td><td>9.7803</td></tr> <tr> <td>6</td><td>9.2197</td></tr> <tr> <td>7</td><td>8.6859</td></tr> </tbody> </table> $\therefore n \geq 7$ Hence, the number of years the machine is in operation is 7.	n	a_n	...	...	5	9.7803	6	9.2197	7	8.6859
n	a_n										
...	...										
5	9.7803										
6	9.2197										
7	8.6859										
12 (a)(i)	$\int \frac{x}{\sqrt{x-1}} dx = \int \frac{x-1+1}{\sqrt{x-1}} dx$ $= \int \left(\sqrt{x-1} + \frac{1}{\sqrt{x-1}} \right) dx = \frac{2}{3}(x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C = \frac{2}{3}(x-1)^{\frac{1}{2}}(x+2) + C$ Alternatively, $u = x \Rightarrow \frac{du}{dx} = 1$; $\frac{dv}{dx} = (x-1)^{-\frac{1}{2}} \Rightarrow v = \frac{(x-1)^{\frac{1}{2}}}{\frac{1}{2}} = 2\sqrt{x-1}$ $\int \frac{x}{\sqrt{x-1}} dx = 2x\sqrt{x-1} - 2 \int \sqrt{x-1} dx = 2x\sqrt{x-1} - 2 \left(\frac{(x-1)^{\frac{3}{2}}}{\frac{3}{2}} \right) + C$ $= 2x\sqrt{x-1} - \frac{4}{3}(x-1)^{\frac{3}{2}} + C = \frac{2}{3}(x-1)^{\frac{1}{2}}(x+2) + C$ NOTE : Integration by substitution with either $u = x-1$ or $u = \sqrt{x-1}$ is also possible.										
(a)(ii)	$\frac{1}{x-2} - \frac{1}{x} - \frac{2}{x^2} = \frac{x^2 - x(x-2) - 2(x-2)}{x^2(x-2)} = \frac{x^2 - x^2 + 2x - 2x + 4}{x^2(x-2)} = \frac{4}{x^2(x-2)}$ $u = 2 + e^{-x} \Rightarrow \frac{du}{dx} = -e^{-x}$. $\int_0^{\ln 4} \frac{4}{(2 + e^{-x})^2} dx = - \int_3^{\frac{9}{4}} \frac{4}{(2 + e^{-x})^2} \frac{1}{e^{-x}} du$ $= \int_{\frac{9}{4}}^3 \frac{4}{u^2(u-2)} du = \int_{\frac{9}{4}}^3 \left(\frac{1}{u-2} - \frac{1}{u} - \frac{2}{u^2} \right) du$ $= \left[\ln u-2 - \ln u + \frac{2}{u} \right]_{\frac{9}{4}}^3$ $= \left(\ln 1 - \ln 3 + \frac{2}{3} \right) - \left(\ln \frac{1}{4} - \ln \frac{9}{4} + \frac{8}{9} \right) = \ln 3 - \frac{2}{9}$										

Q ⁿ	Key Steps / Solution
(b)(i)	 The graph shows the region between the curves $y = \tan x$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$. The region is shaded. Key points labeled are $(0,0)$, $(\frac{\pi}{4}, 1)$, and $(\frac{\pi}{2}, 1)$. The horizontal line $y = 1$ is also shown.
(b)(ii)	Volume required $= \pi \int_0^{\frac{\pi}{4}} (\tan x)^2 dx + \pi(1)^2 \left(\frac{\pi}{4} \right) - \pi \int_0^{\frac{\pi}{2}} (\sin x)^2 dx$ $= \pi \int_0^{\frac{\pi}{4}} ((\sec x)^2 - 1) dx + \frac{\pi^2}{4} - \frac{\pi}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) dx$ $= \pi \left[\tan x - x \right]_0^{\frac{\pi}{4}} + \frac{\pi^2}{4} - \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}}$ $= \pi \left(1 - \frac{\pi}{4} \right) + \frac{\pi^2}{4} - \frac{\pi}{2} \left(\frac{\pi}{2} \right)$ $= \left(\pi - \frac{\pi^2}{4} \right) \text{ unit}^3.$